Yihan Li

liyihan.xyz@gmail.com | +86 18615276438 | liyihan.xyz | github.com/HY-LiYihan

Education

Sun Yat-sen University Sep 2023 – Jul 2027 (Expected)
B.A. in English Language & Literature, School of Foreign Languages *Major GPA: 3.8/4.0*

Minor in Intelligent Unmanned Systems *GPA: 4.0/4.0*
Minor in Computer Science & Technology *GPA: 3.8/4.0*

Publications

Anticipating Contact for Visuo-Tactile Diffusion Policies Under Review
Yating Feng, Yuhang Zheng, Lixin Xu, **Yihan Li**, Jiaju Yin, Renjing Xu

Goldsmith: Gold-Loss-Guided Prompt Training for Agentic Annotation Under Review
Yihan Li, Hanyi Zhang, Xiaoxi Jiang, Man Guo

Unordered Landmark Visual Navigation ECCV 2026
Hao Ren, Junzhe Zhu, **Yihan Li**, Zetong Bi, Le Zheng, Zhi Li, Yiqing Yuan, Zhaoliang Wan, Lu Qi, Hui Cheng

OmniTrav: A Dataset and Benchmark for 360° Vision-based Traversability via Foundation Models IROS 2026
Yihan Li, Hao Ren, Zhenglan Jiang, Junzhe Zhu, Hui Cheng

RAPID Hand: A Robust, Affordable, Perception-Integrated, Dexterous Manipulation Platform for Generalist Robot Autonomy NeurIPS 2025
Zhaoliang Wan, Zetong Bi, Zida Zhou, Hao Ren, Yiming Zeng, **Yihan Li**, Lu Qi, Xu Yang, Ming-Hsuan Yang, Hui Cheng

Work Experience

HCL Lab (HKUST(GZ)) Jan 2026 – Present
Research Assistant *Prof. Renjing Xu*

- Researching tactile sensors and vision models for dexterous manipulation, including multimodal perception pipelines for visuo-tactile policy learning.
- Fine-tuning VLA models and developing VLA-based online RL workflows, with Real2Sim scene reproduction for reproducible evaluation.

Yun Drone Jun 2025 – Mar 2026
UAV Development Intern

- Selected UAV propulsion components and participated in formation-flight development, navigation and localization, and flight-test workflow design.
- Explored RL-based disturbance-resilient control and analyzed flight logs to support anti-disturbance tuning and system iteration.

RAPID Lab (SYSU) Nov 2024 – Jun 2025
Research Assistant *Prof. Hui Cheng*

- Developed the dexterous-hand control stack, visuo-tactile temporal alignment, and MuJoCo Sim2Real validation workflow for RAPID Hand manipulation experiments.
- Contributed to OmniTrav by building the C2T pipeline, mapping pinhole, fisheye, and panoramic RGB inputs into planner-ready 360-degree radial clearance bounds.

RoboTech Association (SYSU) Sep 2024 – Present
President; Competition Team Captain

- Led 160+ members and competition teams, coordinating training schedules, technical reviews, and cross-group collaboration.
- Authored robotics onboarding tutorials across mechanical design, hardware, embedded control, and algorithms.

Projects

Venom VNV: ROS 2 Unified Robotics Platform Mar 2025 – Jul 2025
Project Lead; ROS 2 Platform Developer

- Architected a ROS 2 Humble platform spanning drivers, perception, localization, planning, mission control, simulation, and interface specifications for UGV/UAV/USV scenarios.
- Integrated Livox MID360, Hikvision cameras, AgileX chassis/arm/PX4 drivers with Point-LIO/Fast-LIO, Nav2 TEB, Ego Planner Swarm, YOLO detection, auto-aim tracking, and waypoint task management.

RAPID Hand: Perception-Integrated Dexterous Manipulation Platform Nov 2024 – Jun 2025
Research Assistant

- Co-developed a compact 20-DoF dexterous hand ontology and high-DoF teleoperation/data-collection stack for contact-rich whole-hand manipulation.
- Integrated hardware-level whole-hand perception, synchronizing wrist vision, fingertip tactile sensing, and proprioception with sub-7 ms latency for policy learning.

OmniTrav: 360° **Vision-based Traversability Benchmark** Jun 2025 – Mar 2026
First Author

- Proposed the C2T pipeline and built OmniTrav, a 4.7k-scene, 250k-frame, 577 GB dataset mapping pinhole, fisheye, and panoramic RGB to planner-ready 360-degree radial clearance.
- Designed an automated foundation-model annotation pipeline and a DINOv3-FPN baseline with a cross-attention angle projector for radial distance and obstacle prediction.

GenreShift and Rosetta: LLM Systems for Linguistic Research Oct 2024 – Nov 2025
Project Leader

- Built GenreShift for genre and discipline analysis, extracting quantitative linguistic features such as TTR, MTLD, academic vocabulary, and cross-genre comparisons for LoRA/MoE-based genre transformation.
- Developed Rosetta as an agentic annotation pipeline with concept bootstrap, prompt optimization, embedding-based top-k retrieval, LLM judge routing, review feedback, and Prodigy-compatible JSONL export.

Awards

China Robot and Artificial Intelligence Competition 2025
National First Prize (Team Captain)

National Intelligent Unmanned Systems Application Challenge 2025
National Champion (Team Captain)

National Undergraduate Electronics Design Contest 2025
National Second Prize (Team Captain)

National Undergraduate Robotics Competition - RoboMaster University League 2025
National Second Prize (Vision Group Leader)

Skills

Programming:	ROS1/2, Python, C/C++, MATLAB, STM32/Arduino.
Robotics/ML:	Visual navigation, 3D perception, dexterous manipulation, MuJoCo, TensorRT, OpenVINO, LLM annotation systems.
Engineering and Languages:	Fusion 360, SolidWorks, EDA design; AOPA Multi-rotor VLOS Pilot; Mandarin (native), English (TEM-4), French.